

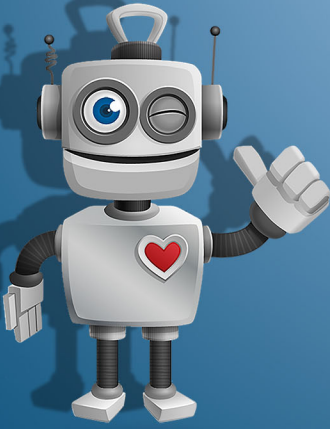
تقدم لكم اللجنة الاكاديمية في فريق

Mechatronics spectrum

امتحان سابق لمادة

الاحكامات المنطقية المبرمجة

الفصل الأول ٢٠١٦



سكند



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Student Name: _____

Number: _____

Q1) Circle the most accurate answer for the following statements (10 marks)

Problems statements 1 and 2 refer to the given ladder program in fig. 1.

1. Decide whether each of these statements is True (T) or False (F).

When activating the input I:010/01:

- (i) An input to I:010/02 gives an output from O:010/00. ~~F~~
 (ii) An input to I:010/03 gives an output from O:010/01. ~~F~~

- A (i) T (ii) T
 B (i) T (ii) F
 C (i) F (ii) T
 D (i) F (ii) F

2. Decide whether each of these statements is True (T) or False (F).

When there is no input to I:010/01:

- (i) An input to I:010/02 gives no output from O:010/00. ~~T~~
 (ii) An input to I:010/04 gives no output from O:010/02. ~~T~~

- A (i) T (ii) T
 B (i) T (ii) F
 C (i) F (ii) T
 D (i) F (ii) F

Problems statements 3 and 4 refer to the given ladder program in fig. 2.

3. Decide whether each of these statements is True (T) or False (F).

The timer in the Figure starts when:

- (i) There is an output. ~~F~~
 (ii) The input ceases (deactivated). ~~T~~

- A (i) T (ii) T
 B (i) T (ii) F
 C (i) F (ii) T
 D (i) F (ii) F

4. Decide whether each of these statements is True (T) or False (F).

When there is an input to In 1(activated), the output is switched:

- (i) On for the time for which the timer was preset.
 (ii) Off for the time for which the timer was preset.

- A (i) T (ii) T
 B (i) T (ii) F
 C (i) F (ii) T
 D (i) F (ii) F

5. Data transfer might be to move

- a. a preset value to output ~~✓~~
 b. data from an input to a register ~~✓~~
 c. data from an output to an actuator ~~✓~~
 d. None

6. One of the following steps is not true when tracking down errors in a PLC.

- a. Have a look and check if there are visible problems.
 b. Check the indicating lights on I/O cards.
 c. Disconnect the PLC power supply and check the system.
 d. None

7. The length of the PLC program depends on

- a. Complexity of the application



Fig (1)

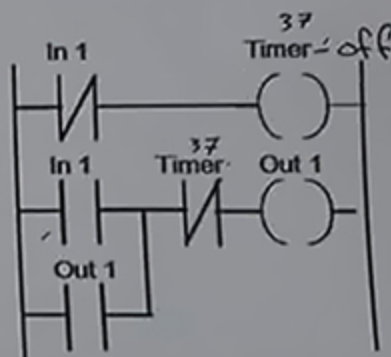


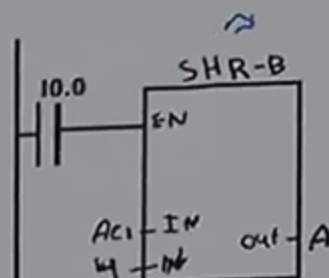
Fig (2)

- b. Type of the PLC
c. Speed of the processor
d. None

8. In a PLC system, we get an overflow when

- a. Multiplying by zero. ✓
b. a result of a math operation is less than the value that could be stored in a memory location.
c. a & b.
d. None

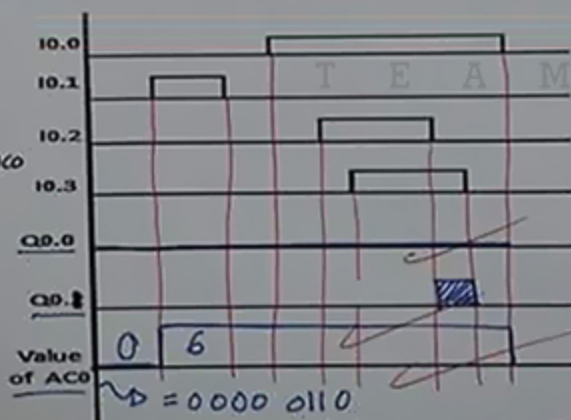
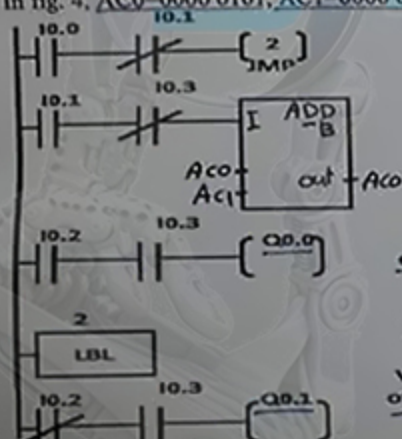
9. For the ladder diagram shown below, if AC1 content is: 1111 1110, the content of AC0 after implementing the program (i.e. I0.0 is activated) is:
a. 0000 0000
b. 0000 1111
c. 1111 1110
d. None



10. A PLC with a long cycle time is suitable for:

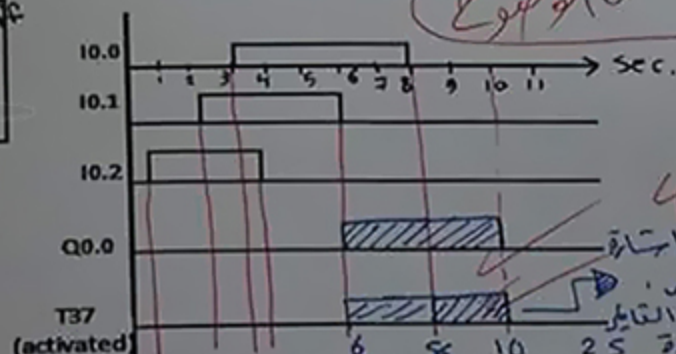
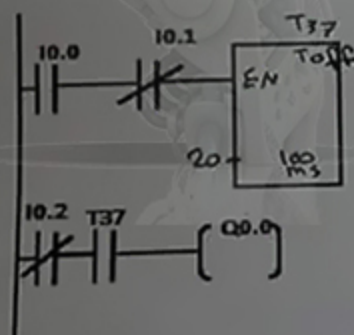
- (i) Short duration inputs.
(ii) High frequency inputs. ✓
A. (i) True (ii) True
B. (i) True (ii) False
C. (i) False (ii) True
D. (i) False (ii) False

Q2) (10 marks) Draw the output for the following Ladder diagrams.
In fig. 4, AC0=0000 0101, AC1=0000 0001 before implementing the ladder.



Fig(4)

$$\begin{array}{r} 0000\ 0101 \\ 0000\ 0001 \\ \hline 0000\ 0110 \end{array}$$



Fig(5)

* عند 6s تقبل إشارة
* EN إشارة التايمر
* عند 8s يبدأ التايمر
بالعد لمدة 2s

1	2	3	4	5	6	7	8	9	10
D	A	C	B	B	C	A	D	B	D